

ITEACH Project: Exotic Options

FIN 5401: Investments

Clark University

November 23, 2021

By: Abby Souza, Lotika Das, & Yuze Ni

Table of Contents

	Page #
Introduction to Exotic Options.....	5
·What are Exotic Options?.....	5
·History.....	5
·Characteristics.....	5
·Difference Between Exotic & Traditional Options.....	6
Binary/Digital/All-or-Nothing Options.....	6
·Definition & Features.....	6
·Comparison of Binary Options & Vanilla Options.....	7
·Cash-or-Nothing Binary Options.....	8
·Asset-or-Nothing Binary Options.....	8
·Pricing Binary Options Using the Black-Scholes Model.....	9
·Profit/Payoff of Binary Options.....	10
Basket Options.....	11
·Definition & Features.....	11
·Who uses Basket Options?.....	12
·Example of a Currency Basket option.....	13
·Payoff of Basket Options.....	13
·Pricing Basket Options.....	14

·A More Accurate & Equally Fast Approximation.....	16
·Pricing Basket Options Using the Monte Carlo Model.....	16
Barrier Options.....	17
·Definition & Features.....	17
·Knock-in Barrier Options.....	18
·Knock-out Barrier Options.....	18
·Other Types of Barrier Options.....	19
·Why Use Barrier Options?.....	19
Other Exotic Options.....	19
·Chooser Options.....	20
·Compound Options.....	20
·Bermuda Options.....	20
·Quantity-Adjusting Options.....	20
·Look-Back Options.....	20
·Extendible Options.....	21
·Range Options.....	21
·Asian Options.....	21
References.....	22
Supporting Appendices.....	24
·Figure 1. Binary Call & Binary Put Option.....	24

·Figure 2. Binary Call Option vs. Vanilla Call Option.....	24
·Figure 3. Cash-or-Nothing Call Option.....	25
·Figure 4. Cash-or-Nothing Put Option.....	25
·Figure 5 Asset-or-Nothing Call Option.....	26
·Figure 6. Asset-or-Nothing Put Option.....	26
·Figure 7. Profit/Payoff of Binary Call Option.....	27
·Figure 8. Profit/Payoff of Binary Put Option.....	27
·Figure 9. Example of a Currency Basket Option.....	28
·Figure 10. Pricing Basket Options.....	28
·Figure 11. Payoff of Basket Option.....	29
·Figure 12. Basket Option vs. Two Vanilla Options.....	29
·Figure 13. Barrier Options: Knock-in Call.....	30
·Figure 14. Barrier Options: Knock-Out Call.....	30

Introduction to Exotic Options

What are Exotic Options?

Exotic options are options that have features that make them more complex than commonly traded vanilla options. Unlike commonly traded vanilla options, exotic options have different payout structures, expiration dates, and strike prices. They may have several triggers relating to determination of payoff. An exotic option may also include a non-standard underlying instrument, developed for a particular client or for a particular market. Exotic options are more complex than options that trade on an exchange and are generally traded over the counter (OTC).

History

The term "exotic option" was popularized by Mark Rubinstein's 1990 working paper (published 1992, with Eric Reiner) "Exotic Options," with the term based either on exotic wagers in horse racing, or due to the use of international terms such as "Asian option," suggesting the "exotic Orient" (Rubinstein, 1990).

Journalist Brian Palmer used the "successful \$1 bet on the superfecta" in the 2010 Kentucky Derby that "paid a whopping \$101,284.60" as an example of the controversial high-risk, high-payout exotic bets that were observed by track-watchers since the 1970s in his article about why we use the term exotic for certain types of financial instrument. Palmer compared these horse racing bets to the controversial emerging exotic financial instruments that concerned then-chairman of the Federal Reserve Paul Volcker in 1980. He argued that just as the exotic wagers survived the media controversy, so will the exotic options (Palmer, 2010).

In 1987, Bankers Trust's Mark Standish and David Spaighton were in Tokyo on business when "they developed the first commercially used pricing formula for options linked to the average price of crude oil." They called this exotic option the Asian option because they were in Asia.

Characteristics

There are two general types of exotic options: path-independent and path-dependent. An option is path-independent if its value depends only on the final price of the underlying instrument. Path-dependent options depend not only on the final price of the underlying instrument, but also on all the prices leading to the final price. An exotic option could have one or more of the following features:

- The payout at maturity depends not just on the value of the underlying instrument at maturity, but its value at several times during the contract's life.
- It could be an asian option depending on some average, a lookback option depending on the maximum or minimum, a barrier option, which ceases to exist if a certain level is reached or not reached by the underlying asset price, a digital option, peroni options, range options, spread options, etc.
- It could depend on more than one index such as in basket options, outperformance options, Himalaya options, or other mountain range options. The manner of settlement

may vary depending on the moneyness of the option at expiry, such as cash or a share option.

- There could be callability and putability rights.
- It could involve foreign exchange rates in various ways, such as a quanto or composite option.
- Even products traded actively in the market can have the characteristics of exotic options, such as convertible bonds, whose valuation can depend on the price and volatility of the underlying equity, the credit rating, the level and volatility of interest rates, and the correlations between these factors.

Difference between Exotic Options and Traditional Options

An exotic option can vary in terms of how the payoff is determined and when the option can be exercised. These options are generally more complex than plain vanilla call and put options. Exotic options usually trade in the over-the-counter (OTC) market. The OTC marketplace is a dealer-broker network, as opposed to a large exchange such as the New York Stock Exchange (NYSE). Further, the underlying asset for an exotic can differ greatly from that of a regular option. Exotic options can be used in trading commodities such as lumber, corn, oil, and natural gas as well as equities, bonds, and foreign exchange. Speculative investors can even bet on the weather or price direction of an asset using a binary option.

Despite their embedded complexities, exotic options have certain advantages and disadvantages over traditional options, which include:

- Pros
 - Customized to specific risk-management needs of investors
 - A wide variety of investment products to meet investors' portfolio needs
 - In some cases, lower premiums than regular options
 - Exotic options can help offset risk in a portfolio
- Cons
 - Some exotic options can have increased costs given their added features
 - Exotic options do not guarantee a profit
 - The reaction of price moves for exotics to market events can be different than traditional options

Binary/Digital/All-or-Nothing Options

Definition & Features

Binary options, also known as digital or all-or-nothing options, are a type of exotic option where those involved in the transaction are assigned one of two outcomes based on whether or not the option expires in the money. These options depend on an outcome of “yes” or “no,” which is why they are called binary options. If the binary option expires in the money, traders receive a payout. However, if the binary option expires out of the money, these traders experience a loss. The fixed payout and loss amount are set by the binary option, and these options do not allow

traders to take a position in the underlying security. Most of these exotic options also trade outside the United States (Downey, 2021). The two most popular types of binary options are cash-or-nothing options and asset-or-nothing options.

Regarding how binary options actually work, all binary options have an expiration date or time. At this time, the price of the underlying asset of the option must be on the “correct” side of the strike price (K) for the trader to make money. The “correct” side of the strike price is based on the trade taken. Once the binary option expires, the option is exercised automatically. This means that the gain or loss from the option trade is added or subtracted automatically from the trader’s account, meaning the buyer of the option will either receive an option payout or lose their entire investment on the trade. With these options, there is no in between. On the seller side, the seller of the option will either earn the buyer’s premium or be required to make a full payout (Downey, 2021). Due to the nature of how binary options work, they are very prone to fraud and are banned by regulators in many jurisdictions, as they are seen as a form of gambling. In fact, in the United States, the FBI is investigating binary option scams throughout the world (Wikipedia, 2021).

Binary options can be extremely simple. For example, it could be as simple as whether or not Apple’s stock price is above \$153 on November 24th at 11:30am. The trader’s only options are yes, the stock price will be higher than \$153 or no, the stock price will be lower than \$153. To take it a step further, say the trader believes that Apple’s stock will be \$153 on this date and time and is willing to wager \$100 on this trade. If Apple’s shares trade above \$153 at that specific date and time, the trader will receive a payout based on the previously agreed upon terms. So, if the agreed upon payout was 80%, the trader would receive \$80. If the shares trade below \$153, the trader would lose their entire \$100 investment (Downey, 2021). To see a binary call option and a binary put option displayed graphically, please see Figure 1 in our supporting appendices.

Comparison of Binary Options & Vanilla Options

Next, we will compare binary options to the most basic type of options, vanilla options. Vanilla options can be broken down into American options and European options. An American option gives the option holder the right to buy or sell an underlying asset at a specific price on or before the option expiration date. A European option is the same as an American option, but the option right can only be exercised on the expiration date. A comparison of binary options and vanilla options is as follows:

- Binary options do not provide the chance to take a position in the underlying asset, while vanilla options do.
- With vanilla options, traders experience fixed risk; however, profits vary based on how far the price of the underlying asset moves (Downey, 2021).
- Binary options usually specify a fixed maximum payout, while the maximum risk is limited in that it is equal to the amount invested into the option.
- Unlike vanilla options, movement of the underlying asset’s price does not affect the payout or loss of a binary option.

- Some binary options also can be closed before expiration, similar to an American option but different from a European option. However, closing a binary option before expiration could potentially limit the payout.
- Vanilla options trade on U.S. exchanges regulated by the Securities Exchange Commission (SEC) and other agencies and are subject to United States option market regulation. On the other hand, most binary options trade outside the United States and may not be regulated. This means that many binary option brokers do not have to meet a specific standard, so investors should be careful of fraud (Downey, 2021).

A graphic comparison of a binary call option and a vanilla call option can be seen in Figure 2 in our supporting appendices.

Cash-Or-Nothing Binary Options

A cash-or-nothing option is one of the most common binary options, which can be broken down into a cash-or-nothing call and a cash-or-nothing put. A cash-or-nothing call pays nothing if the asset price drops below the strike price at time T and pays a fixed amount, represented by Q , if the asset price ends up above the strike price. If we consider a risk-neutral world, the probability that the asset price will be above the strike price at the option's maturity is equal to $N(d_2)$. This means that the value of a cash-or-nothing call is equal to $Qe^{-rT}N(d_2)$. On the other hand, a cash-or-nothing put is opposite to a cash-or-nothing call in that it pays Q if the asset price is below the strike price and pays nothing if the asset price is above the strike price. This means that the value of the cash-or-nothing put is $Qe^{-rT}N(-d_2)$. A vanilla European call option is equivalent to a short position in a cash-or-nothing call and a vanilla European put option is equivalent to a long position in a cash-or-nothing put (Topic 2, Reference 2). It is important to note that with cash-or-nothing options, these options pay in full if a condition is met or they pay nothing if the condition is not met; there is no in-between. Further, as its name suggests, these types of options pay out in cash. Since the payout is fixed, it does not matter how deep in the money the option is. All that matters is if the underlying asset closes above the strike price for a call option or below the strike price for a put option (Scott, 2021). A cash-or-nothing call option and a cash-or-nothing put option are expressed graphically in Figure 3 and Figure 4 in the supporting appendices.

In terms of an example, suppose that Apple currently trades at \$150 at 1pm on November 1st. A trader believes that Apple will trade above \$160 before the end of the trading day. For this reason, the trader purchases 10 Apple \$150 cash-or-nothing options at 1pm for \$50 a contract. If Apple closes above \$150 at the end of the trading day on November 1st, the trader would receive a profit of \$50 per contract. However, if Apple closes below \$150 on this date, the trader loses its entire investment which would be \$500 (10 options * \$50 per option contract).

Asset-Or-Nothing Binary Options

An asset-or-nothing option is another popular binary option. Like all options, this binary option can be separated into an asset-or-nothing call and an asset-or-nothing put. An asset-or-nothing call pays nothing if the underlying asset price is below the strike price. On the other hand, this option pays the asset price if it is above the strike price. This means that the value of an asset-or-

nothing call is equal to $S_0 e^{-qT} N(d_1)$. For an asset-or-nothing put, the put option pays nothing if the underlying asset price is above the strike price and pays the asset price if the underlying asset price is below the strike price. The value of an asset-or-nothing put option is equal to $S_0 e^{-qT} N(-d_1)$. A vanilla European call option is equivalent to a long position in an asset-or-nothing call and a vanilla European put option is equivalent to a short position in an asset-or-nothing put (Topic 2, Reference 2). It is important to note that asset-or-nothing options either pay or do not pay a fixed amount, depending on whether they expire in the money or not. These binary/digital options are often used as an effective hedging tool because of their simplified risk and payout structure. Additionally, as its name suggests, an asset-or-nothing option pays with the physical delivery of the underlying asset. Since these options are digital or binary, they pay a predetermined payout, or they pay zero (Scott, 2021). An asset-or-nothing call option and an asset-or-nothing put option are expressed graphically in Figure 5 and Figure 6 in the supporting appendices.

Regarding an example of an asset-or-nothing call option, suppose that gold trades at \$1,260 per ounce at 1pm on November 1st. A trader believes that gold will trade above \$1,275 per ounce before the end of the trading day on November 1st. The trader purchases 10 gold \$1,275 asset-or-nothing call options at 1pm. If gold closes above \$1,275 at the end of the trading day, the trader will receive 10 contracts worth of gold. If gold does not close above \$1,275 per ounce, the trader will lose their entire investment (Scott, 2021).

Pricing Binary Options Using the Black-Scholes Model

The Black-Scholes Model is a differential equation that is widely used to price options. This model requires five different input variables including the strike price, the current asset price, the time to expiration, the risk-free rate, and the volatility. The model predicts that the price of heavily traded assets follows a geometric Brownian motion with constant drift and volatility. Additionally, this model makes six certain assumptions. The first assumption is that no dividends are paid out during the life of the option. The second assumption is that markets are random, so market movements cannot be predicted. The third assumption of this model is that there are no transaction costs in buying the option. The fourth assumption is that the risk-free rate and volatility of the underlying asset are known and are constant. The fifth assumption is that the returns on the underlying asset are log-normally distributed. Finally, the sixth assumption is that the option is a European option and it can only be exercised at expiration (Hayes, 2021).

While the Black-Scholes Model is typically only applied to European options, it can be interpreted to apply to binary options. The Black-Scholes call option formula is calculated by multiplying the asset price by the cumulative standard normal probability distribution function. After that, the net present value of the strike price multiplied by the cumulative standard normal probability distribution is subtracted from the resulting value of the previous calculation (Hayes, 2021). The mathematical notation of this model is as follows:

Call Option:

$$C(S_t, t) = N(d_1)S_t - N(d_2)Ke^{-r(T-t)}$$

$$d_1 = \frac{1}{\sigma\sqrt{T-t}} \left[\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t) \right] \quad \text{Put Option:}$$

$$\begin{aligned} P(S_t, t) &\equiv Ke^{-r(T-t)} - S_t + C(S_t, t) \\ &= N(-d_2)Ke^{-r(T-t)} - N(-d_1)S_t \end{aligned}$$

where:

C= call option price

P= put option price

S= underlying asset price

K= strike price

r= risk-free interest rate

t= time to maturity

N=normal distribution

The Black-Scholes formula for the price of a European call option or put option can be interpreted as a binary option by breaking a call option down into an asset-or-nothing call option minus a cash-or-nothing call option, or asset-or-nothing put minus cash-or-nothing put for a put option (Wikipedia, 2021). The value of each of these options is as follows:

$$\text{Asset-or-Nothing Call: } C = Se^{-q(T-t)} N(d_1)$$

$$\text{Asset-or-Nothing Put: } P = Se^{-q(T-t)} N(-d_1)$$

$$\text{Cash-or-Nothing Call: } C = e^{-r(T-t)} N(d_2)$$

$$\text{Cash-or-Nothing Put: } P = e^{-r(T-t)} N(-d_2)$$

The Black-Scholes formula is the difference between two terms, and these two terms equal the values of the binary call options (Wikipedia, 2021).

Profit/Payoff of Binary Options

The profit and payoff of binary options can be broken down into a binary option call payoff and a binary option put payoff. In terms of a binary call option, the payoff would be:

$$c^b(T) = \begin{cases} 0 & \text{if } S(T) \leq K \\ 1 & \text{if } S(T) > K \end{cases}$$

For a binary put option, the payoff would be:

$$p^b(T) = \begin{cases} 1 & \text{if } S(T) \leq K \\ 0 & \text{if } S(T) > K \end{cases}$$

Regarding profit, the profit for the option buyer would be equal to the payoff minus the option premium. The profit for the option seller would be the payoff plus the option premium. These profits are the same for a call option and a put option. A profit/payoff plot of a binary call option is represented by Figure 7 and a profit/payoff plot of a binary put option is represented by Figure 8 in the supporting appendices.

Basket Options

Definition & Features

Basket options are contracts based on the weighted value of a basket of goods. It is a type of financial derivative where the underlying asset is a group, or basket of commodities, securities, or currencies. The contract is set up to represent the entire sum of values in the basket, and the individual purchaser would acquire each of the underlying assets if he or she executes the option (Caldana, Fusai, Gnoatto, Grasselli, 2021).

In a basket option, multiple financial products are simplified into one contract and a single maturity price is set for the same. Due to this, basket options are used in order to hedge risks in foreign currencies. In particular, basket options are popular with companies that will need to obtain various currencies at a future date and want to mitigate the chance the exchange will become unfavorable. This exotic option has all the characteristics of a standard option, but with strike prices based on the weighted values of its components.

Currency baskets are the most popular type of basket option, and they will settle in the holder's home currency. Because it involves just one transaction, a basket option often costs less than multiple single options as it saves on commissions and fees. A wide range of baskets are available (e.g. interest rate, FX, equity, or commodity related). Baskets allow for a balanced exposure and are less volatile with respect to an index.(The Financial Engineer, 2021)

The downside of the basket option is that there is limited liquidity for such options, so getting out before expiry may require additional offsetting transactions. However, there are very limited alternatives to this option.(Investopedia, 2021)

One alternative to a basket option is an index option. However, if an investor is seeking exposure to an obscure economic sector or to an emerging country market, then none of the indices that

underlie listed index options is likely to be sufficiently correlated with the portfolio of interest. Thus, there will be significant tracking error risk that can cost more than the additional cost of the basket option.

A second alternative is a collection of individual stock options. This alternative may be expensive due to transaction costs and because the implied volatility of each option includes a large component of specific stock risk.

A basket is defined as the weighted arithmetic average of the n stock prices $S_1(t), \dots, S_n(t)$ with a given vector of weights $w = (w_1, \dots, w_n) \in \mathbb{R}^n$ at time T :

$$A_n(T) = \sum_{k=1}^n w_k S_k(T).$$

We assume, without loss of generality, that $k = 1$. A basket call option gives the holder the right, but not the obligation to purchase the portfolio of assets at a fixed price, K , known as the option's strike price (Caldana, Fusai, Gnoatto, Grasselli, 2021).

Who uses Basket Options?

Investors who are inclined toward having a wide exposure to specific regions to have a protection against an unfavorable shift in a targeted index buy basket options. Additionally, multinational companies like Subway or McDonalds use basket options to hedge against foreign exchange rate risk, as they have businesses in multiple countries. Basket options can be used for trading in the following ways:

- Basket options are used by multinational corporations to hedge against the foreign exchange risk rate. Even the investors who need to broaden their exposure to a particular region buy basket options.
- A substitute to a basket option is an index option. However, if an investor is searching for exposure to an obscure economic sector, then none of the listed index options is likely to be adequately correlated with the portfolio of interest. Thus, there will be significant tracking error risk that can cost more in the basket option.
- Another alternative is a multitude of individual stock options. This alternative may cost more due to transaction costs and because the implied volatility of each option has a large component of specific stock risk. (Moksha Gala, 2021)

The benefits of basket options include:

- An option on a portfolio of assets is lower priced than an option on each of the portfolios. The lesser the correlation between the various assets that make up the basket, the greater is the cost-saving.
- Baskets are available for interest rate, FX, equity, or commodity-related.
- Baskets allow its client for a balanced exposure and are less unstable to an index (Mitchell, 2021)

Example of a Currency Basket option

Suppose an international US-based company is willing to buy a basket option on the Singapore dollar and New Zealand dollar. In this circumstance, if the Singapore dollar or New Zealand dollar currency value falls against the US dollar, then they would want to sell at a specified price point so that they can avoid further depreciation. The company would buy a put with a basket containing the SGD to USD and NZD to USD.

Suppose the company knows they have more exposure to SGD and NZD, therefore they opt for a weight of 60/40. Based on current rates, at the time of the transaction, an index value can be created.

Let's assume that the SGD/USD rate is 0.74, and the NZD/USD rate is 0.70.

The index value is: $(0.74 \times 0.6) + (0.70 \times 0.4) = 0.444 + 0.280 = 0.724$

In this scenario, suppose the buyer of the put wants a strike price at 0.71 for the basket, and the option will expire in one year. The buyer and the seller will determine the terms of the contract, will agree on the amount, will determine the amount of currency for the option as well as a premium.

Market changes shape every day and if SGD and NZD rise over the next year and the index pushes to 0.75, the buyer will not exercise the option because if the index value is more than the strike price, it will be a loss for the investor if he uses the option. In this case, the buyer will lose their premium and the seller will be able to keep the premium as a profit. However, if the currencies SGD and NZD falls and becomes 0.680 then the buyer will exercise the put option, as he will be able to sell the basket at 0.71, since the weighted value of the currencies is only 0.680.

The difference between the strike price and the weighted value, less the premium, is the buyer's profit. The seller's loss is the weighted value minus the strike price, plus the premium received. The risky currencies have different weights in the basket to reflect the details of the exposure. For example, a basket call on two currencies US-\$ and JP-¥ pays off at maturity T, where $S_1(t)$ denotes the exchange rate of €/€ and $S_2(t)$ denotes the exchange rate of €/¥ at time t, a_i the corresponding weights and K the strike (Mitchell, 2021).

$$\max\left(a_1 \frac{S_1(T)}{S_1(0)} + a_2 \frac{S_2(T)}{S_2(0)} - K; 0\right)$$

Payoff of Basket Options

The payoff for a basket call option is:

$$\text{Payoff}_c(T) = N * P * \max(R - K, 0)$$

And the payoff for basket put option is

$$\text{Payoff}_p(T) = N * P * \max(K - R, 0)$$

Where, $R = \sum_{i=1}^n w_i S_i / F_i$ the weighted average of the basket return

N the notional amount

P the option participation rate

w_i the weight for asset i , $i = 1, \dots, n$

F_i the initial Fixing for asset i , $i = 1, \dots, n$

K the basket percentage strike

S_i the spot price for asset i at time T

Pricing Basket Options

One way to price basket options is to set the price in a consistent way with plain vanilla options. Therefore, the basic model assumption is a log-normal process for the individual correlated basket components. A disintegration into uncorrelated components of the exchange rate processes is the basis for pricing.

$$dS_i = \mu_i S_i dt + S_i \sum_{j=1}^N \Omega_{ij} dW_j$$

Here, μ_i denotes the difference between the foreign and the domestic interest rate of the i -th currency pair, dW_j the j -th component of independent Brownian increments. The covariance matrix is given by $C_{ij} = (\Omega \Omega^T)_{ij} = \rho_{ij} \sigma_i \sigma_j$. Here σ_i denotes the volatility of the i -th currency pair and ρ_{ij} the correlation coefficients.

Starting with the uncorrelated components the pricing problem is reduced to the N -dimensional integration of the payoff. This method is accurate but rather slow for more than two or three basket components.

A simple approximation method assumes that the basket spot itself is a log-normal process with drift μ and volatility σ driven by a Wiener Process $W(t)$

$$dS(t) = S(t) [\mu dt + \sigma dW(t)]$$

with solution

$$S(T) = S(t) e^{\sigma W(T-t) + \left(\mu - \frac{1}{2}\sigma^2\right)(T-t)}$$

given we know the spot $S(t)$ at time t . It is a fact that the sum of log-normal processes is not log-normal itself, but as a crude approximation it is certainly a quick method that is easy to implement. In order to price the basket call, the drift and the volatility of the basket spot need to be determined. This is done by matching the first and second moment of the basket spot with the first and second moment of the log-normal model for the basket spot. The moments of log-normal spot are:

$$E(S(T)) = S(t)e^{\mu(T-t)}$$

$$E(S(T)^2) = S(t)^2 e^{(2\mu + \sigma^2)(T-t)}$$

We solve these equations for the drift and volatility:

$$\mu = \frac{1}{T-t} \ln \left(\frac{E(S(T))}{S(t)} \right)$$

$$\sigma = \sqrt{\frac{1}{T-t} \ln \left(\frac{E(S(T)^2)}{E(S(T))^2} \right)}$$

In these formulas, we now use the moments for the basket spot:

$$E(S(T)) = \sum_{i=1}^N \alpha_i S_i(t) e^{\mu_i(T-t)}$$

$$E(S(T)^2) = \sum_{i,j=1}^N \alpha_i \alpha_j S_i(t) S_j(t) e^{(\mu_i + \mu_j + \sum_{k=1}^N \Omega_{ki} \Omega_{jk})(T-t)}$$

The pricing formula is the well-known Black-Scholes-Merton formula for plain vanilla call options:

$$v(0) = e^{-r_d T} (F N(d_+) - K N(d_-))$$

$$F = S(0) e^{\mu T}$$

$$d_{\pm} = \frac{1}{\sigma \sqrt{T}} \left(\ln \left(\frac{F}{K} \right) \pm \frac{1}{2} \sigma^2 T \right)$$

Here, N denotes the cumulative normal distribution function and r_d the domestic interest rate.

A More Accurate and Equally Fast Approximation

The previous approach can be taken one step further by introducing one more term in the Itô-Taylor expansion of the basket spot, which results in

$$\begin{aligned}
v(0) &= e^{-r_d T} (F \mathbf{N}(d_1) - K \mathbf{N}(d_2)) \\
F &= \frac{S(0)}{\sqrt{1-\lambda T}} e^{\left(\mu \frac{\lambda}{2} + \frac{\lambda \sigma^2}{2(1-\lambda T)} \right) T} \\
d_2 &= \frac{\sigma \sqrt{1-\lambda T} \left(\left(1 + \frac{\lambda}{1-\lambda T} \right) \sigma^2 T - 2 \ln \frac{F \sqrt{1-\lambda T}}{K} \right)}{\sqrt{T} \lambda} \\
d_1 &= \sqrt{1-\lambda T} d_2 + \frac{\sigma \sqrt{T}}{\sqrt{1-\lambda T}}
\end{aligned}$$

The new parameter λ is determined by matching the third moment of the basket spot and the model spot. Most remarkably this major improvement in the accuracy only requires a marginal additional computation effort.

Pricing Basket Options Using the Monte Carlo Method

This is the most popular method used to price a Basket option. Monte Carlo methods, or Monte Carlo experiments, are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. The underlying concept is to use randomness to solve problems that might be deterministic in principle. They are often used in physical and mathematical problems and are most useful when it is difficult or impossible to use other approaches. Monte Carlo methods are mainly used in three problem classes: optimization, numerical integration, and generating draws from a probability distribution.

Theorem: Let P be a function of the solution of one or more SDEs, and P_h the discrete approximation using a timestep $h_l = 2^{-l} T$. If there exist independent estimators $\hat{P}_l$ based on N_l Monte Carlo samples, with computational complexity (cost) C_l , and positive constants $\alpha \geq \frac{1}{2}$, β , c_1 , c_2 , c_3 such that

$$\begin{aligned}
i) \quad & \left| \mathbb{E}[\hat{P}_l] - P \right| \leq c_1 h_l^\alpha \\
ii) \quad & \mathbb{E}[\hat{Y}_l] = \begin{cases} \mathbb{E}[\hat{P}_0], & l = 0 \\ \mathbb{E}[\hat{P}_l - \hat{P}_{l-1}], & l > 0 \end{cases} \\
iii) \quad & \mathbb{V}[\hat{Y}_l] \leq c_2 N_l^{-1} h_l^\beta \\
iv) \quad & C_l \leq c_3 N_l h_l^{-1}
\end{aligned}$$

then there exists a positive constant c_4 such that for any $\varepsilon < e^{-1}$ there are values L and N_l for which the multilevel estimator:

$$\hat{Y} = \sum_{l=0}^L \hat{Y}_l,$$

$$\text{has Mean Square Error } MSE \equiv \mathbb{E} \left[\left(\hat{Y} - \mathbb{E}[P] \right)^2 \right] < \varepsilon^2$$

with a computational complexity C with bound

$$C \leq \begin{cases} c_4 \varepsilon^{-2}, & \beta > 1, \\ c_4 \varepsilon^{-2} (\log \varepsilon)^2, & \beta = 1, \\ c_4 \varepsilon^{-2-(1-\beta)/\alpha}, & 0 < \beta < 1. \end{cases}$$

Barrier Options

Definition & Features

Barrier options are another popular type of exotic option. A barrier option is a type of derivative where the payoff of the option depends on whether or not the underlying asset has reached or has gone beyond a previously determined price. A barrier option can be considered “knock-out,” meaning that it is worthless at expiration if the underlying asset exceeds a certain price, which would limit profits for the option holder and limit the loss for the option seller. These options can also be considered “knock-in,” meaning the option has no value until the underlying asset reaches a predetermined price. Barrier options are considered to be exotic options because they are considered to be more complex than American and European vanilla options. These options are also considered to be path-dependent options because their value changes as the underlying asset’s value changes. This means that the payoff of a barrier option is based on the underlying asset’s price path. A barrier option will become worthless or will be activated based on the crossing of a price point barrier (Chen, 2021).

Knock-In Barrier Options

As briefly mentioned above, knock-in barrier options are options where the rights associated with the option only come into existence when the price of the underlying asset reaches a specific barrier during the option’s life. Once this barrier is “knocked-in,” meaning it comes into existence, the option will remain in existence until it reaches its expiration. Knock-in options are further classified as “up-and-in” or “down-and-in.” When they are up-and-in, the option only

comes into existence if the price of the underlying asset is greater than the predetermined barrier, which is set above the underlying asset's initial price. When the option is down-and-in, it only comes into existence when the underlying asset price decreases below a predetermined barrier that is set below the underlying asset's initial price (Chen, 2021). A knock-in call option can be seen graphically in Figure 13 of the supporting appendices below.

Regarding an example of knock-in barrier options, suppose an investor purchases an up-and-in call option with a strike price of \$50 and a barrier of \$55 when the underlying stock is trading at \$45. The option will not move into existence until the underlying stock price increases above \$55. The option only becomes useful if the underlying stock price reaches \$55. If it does not, the option is never exercised, and the option buyer loses what they paid for the option (Chen, 2021).

Knock-Out Barrier Options

Opposite of knock-in barrier options, knock-out barrier options do not exist if the underlying asset price reaches the barrier during the option's life. Knock-out barrier options are classified as "up-and-out" or "down-and-out." An up-and-out barrier option does not exist if the underlying asset price moves above the barrier that is set above the underlying asset's initial price. A down-and-out option does not exist if the underlying asset price moves below the barrier that is set below the underlying asset's initial price. If an underlying asset reaches the barrier at any point during the option's life, the option is knocked-out or discontinued (Chen, 2021).

Regarding an example of knock-out barrier options, suppose a trader purchases an up-and-out put option with a barrier of \$35 and a strike price of \$30 when the underlying asset is trading at \$28. If the underlying asset's price rises above \$35 during the life of the option, the option will no longer exist. This means that the option is now worthless, even if it only touched \$35 briefly then dropped back down below \$35 (Chen, 2021). A graphical depiction of a knock-out call option is represented by Figure 14 in the supporting appendices below.

Other Types of Barrier Options

Rebate barrier options allow both knock-in and knock-out options to have provisions to provide rebates to options holders if the barrier option does not meet the barrier price and becomes worthless. Rebates come in the form of a percentage of the option premium paid by the option holder when purchasing the option (Chen, 2021).

Turbo warrant barrier options are down-and-out options that are highly leveraged and are characterized by low volatility. They are mainly traded in Europe and Hong Kong and they are especially popular in Germany where they are used for speculation purposes (Chen, 2021).

Parisian barrier options are options that do not trigger the option contract if the option reaches the barrier price. Rather, the underlying asset's price has to spend a predetermined amount of time beyond the set barrier price for the option contract to kick in. This means that the amount of time the underlying asset's price spends outside or inside the barrier price range is measured with this type of barrier option (Chen, 2021).

Why Use Barrier Options?

- Lower cost than plain vanilla options (due to risk of knocking out, or never knocking in)
- A big attraction is the ability to fine-tune the structure to achieve specific yields/participations (see Bonus Certificates)
- Once the barrier is reached, barrier options have the same benefits as a plain vanilla option
- Leverage and potentially higher returns
- Limited risk (small premium)
- Barrier options can enable investors to take an aggressive view on skews and sell volatility at lower strikes than would normally be liquid (The Financial Engineer, 2021).

Other Exotic Options

While we talked about many of the popular exotic options above, there are many other exotic options that we have not yet talked about. In this section, we will be exploring other exotic options including chooser options, compound options, bermuda options, quantity-adjusting options, look-back options, extendible options, range options, and asian options.

Chooser Options

Chooser options allow investors to choose whether an option is a put or a call option during a certain point in the option's life. With this type of option, the strike price and the expiration are typically the same whether it is a put or a call. These exotic options are usually used by investors when there is a company event that could lead to volatility or price fluctuations in the asset price, such as a product release (Chen, 2021).

Compound Options

Compound options are a type of exotic option that gives owners the right, but not the obligation, to buy another option at a specified price on or by a specified date. Usually, the underlying asset of a call or put option is an equity security. However, with compound options, the underlying

asset is another option. Compound options come in four types including call on call (1), call on put (2), put on put (3), and put on call (4) (Chen, 2021).

Bermuda Options

Bermuda options are exotic options that can be exercised at preset dates in addition to the expiration date. An example of a bermuda option would be only allowing investors to exercise the option on the first of the month. By allowing investors to exercise the option on a preset date in addition to the expiration date, these exotic options provide investors with more control over when the option is exercised, leading to a higher premium compared to European options. However, bermuda options are still cheaper than American options since American options can be exercised at any time (Chen, 2021).

Quantity-Adjusting Options

Quantity-adjusting options, also known as quanto-options, expose the buyer to foreign assets, but also provide the buyer with the safety of a fixed exchange rate in the buyer's home currency. This type of exotic option is perfect for investors who want to gain exposure in foreign markets, but who also may be worried about how exchange rates will trade when it is time for the option to be paid out. However, many times, this type of option will require an option premium that is way beyond what a traditional option would require due to the "two-in-one" package of being exposed to foreign assets but being paid out in the home country's currency (Chen, 2021).

Look-Back Options

Look-back options are a type of exotic option that do not have a fixed exercise price in the beginning. Instead, the strike price of the option resets to the best price of the underlying asset as it changes. This means that the holder of the look-back option can choose the most favorable exercise price retrospectively for the period of the option. For these reasons, look-back options get rid of the risk associated with timing market entry, resulting in them being more expensive than vanilla options. However, these exotic options can be risky when an investor pays the more expensive premium than a vanilla option, but the stock price does not move enough to generate a profit (Chen, 2021).

Extendible Options

Extendible options are exotic options that allow investors to extend the expiration date of the option. As an option gets closer to its expiration, these options have a specific period that the option can be extended. This feature is available to both the option buyer and the option and seller, and it can be especially helpful if the option is not profitable yet or if it is out of the money at its expiration (Chen, 2021).

Range Options

Range options are a type of option that have a payoff based on the difference between the maximum and minimum price of the underlying asset during the life of the option. The benefits

of these options are that they eliminate the risk associated with market entry and exit timing, making them more expensive than vanilla and look-back options (Chen, 2021).

Asian Options

Asian options are exotic options that take the average price of the underlying asset to determine if there is a profit as compared to the strike price. If the average price is less than the strike price at the expiration of the option, the option is worthless (Chen, 2021).

References

- Binary options*. SLCG Blog. (2013, January 14). Retrieved from <https://blog.slcg.com/2013/01/introduction-to-binary-options.html>
- Binary Put Option Explained*. Binary put option explained. (2017). Retrieved from <https://www.theoptionsguide.com/binary-put-option.aspx>.
- Chen, J. (2021, June 23). *Exotic option definition*. Investopedia. Retrieved from <https://www.investopedia.com/terms/e/exoticoption.asp>.
- Chen, J. (2021, August 5). *Barrier Option Definition*. Investopedia. Retrieved from <https://www.investopedia.com/terms/b/barrieroption.asp>.
- Fin Pricing 2021: Basket Option Payoffs / Payoff of Basket Options. Retrieved from <https://finpricing.com/lib/EqBasket.html>
- FX Basket Options Jürgen Hakala, Uwe Wystup (September 2008): Pricing Basket option. Retrieved from <https://papers.ssrn.com/sol3/results.cfm>
- Ganesh Nayak Blog : Decode finance - Who uses Basket Options? Retrieved from <https://blog.fintelligents.com/basket-option/>
- General closed-form basket option pricing bounds: Ruggero Caldana, Gianluca Fusai, Alessandro Gnoatto and Martino Grasselli - Definition & Features. Retrieved from <file:///D:/Classes/Investments/Exotic%20Options/SSRN-id2376134.pdf>
- Hayes, A. (2021, September 2). *What is the Black-Scholes Model?* Investopedia. Retrieved from <https://www.investopedia.com/terms/b/blackscholes.asp>
- Hull, J. (2022). Chapter 13: The Black Scholes Formula. In *Options, futures, and other derivatives*. essay, Pearson.
- Mike Giles: Multilevel Monte Carlo for Basket Options (Dec 13-16, 2009), Pricing Basket option using Monte Carlo Method. Retrieved from <https://papers.ssrn.com/sol3/results.cfm>
- Mitchell, C. (2021, June 28). *Trading binary options in the U.S.: Your guide*. Investopedia. Retrieved from <https://www.investopedia.com/articles/active-trading/061114/guide-trading-binary-options-us.asp>
- Mitchell, C. (September 02, 2021) Investopedia - Basket options, definition & Features. Retrieved from <https://www.investopedia.com/terms/b/basketoption.asp>
- Monte Carlo methods for option pricing : (Updated 9 August 2021) Pricing Basket option using Monte Carlo Method. Retrieved from https://en.wikipedia.org/wiki/Monte_Carlo_methods_for_option_pricing

Palmer, B. (14 July 2010). *Why Do We Call Financial Instruments "Exotic"? Because some of them are from Japan*. Slate. Retrieved 9 September 2013. <https://slate.com/news-and-politics/2010/07/when-did-people-start-referring-to-financial-instruments-as-exotic.html>

Rubinstein, Mark; Reiner, Eric (1995). *Exotic Options*. Working Paper, University of California at Berkeley. <https://ideas.repec.org/p/ucb/calbrf/rpf-220.html>

Seth, S. (2021, September 13). *Arbitrage strategies with binary options*. Investopedia. Retrieved from <https://www.investopedia.com/articles/active-trading/040115/arbitrage-strategies-binary-options.asp>.

Scott, G. (2021, August 24). *Asset-or-nothing call option definition*. Investopedia. Retrieved from <https://www.investopedia.com/terms/a/aonc.asp>

Scott, G. (2021, September 17). *Cash-or-nothing call definition*. Investopedia. Retrieved from <https://www.investopedia.com/terms/c/conc.asp>

The financial engineer - Barrier options: Why use the Barrier Option

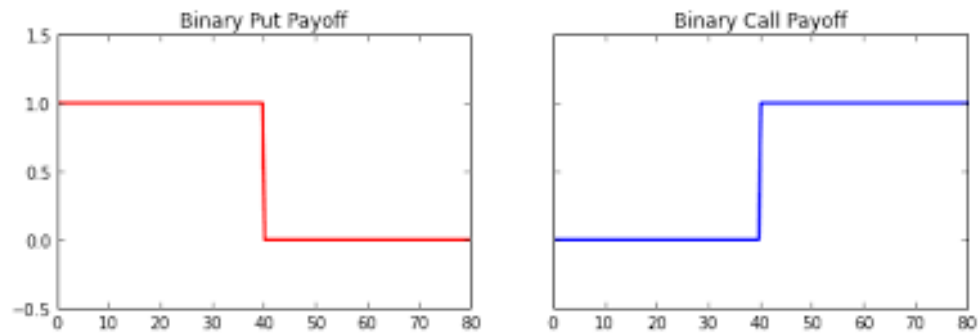
<https://thefinancialengineer.org/options-futures-other-derivatives/exotic-options/barrier-option-2/>

Wikimedia Foundation. (2021, November 1). *Black–Scholes model*. Wikipedia. Retrieved from https://en.wikipedia.org/wiki/Black%E2%80%93Scholes_model#Black%E2%80%93Scholes_formula.

William Falloon; David Turner, eds. (1999). *The evolution of a market*. Managing Energy Price Risk. London: Risk Books.

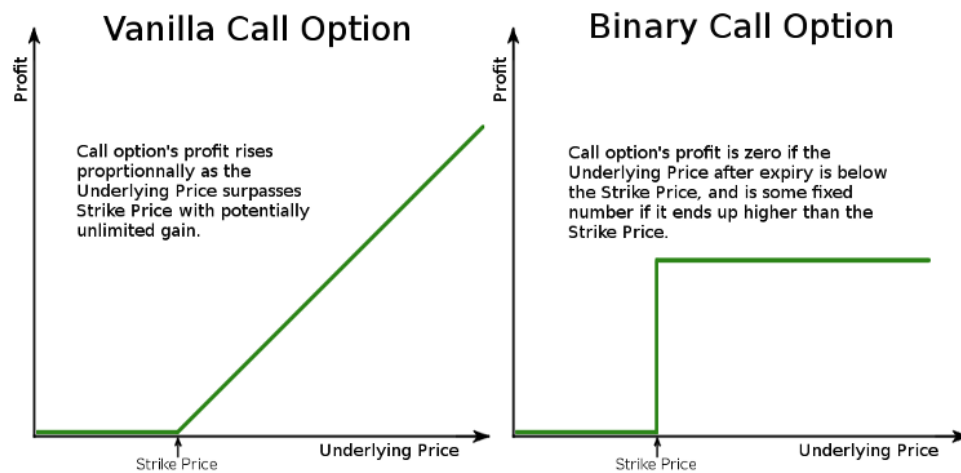
Supporting Appendices

Figure 1. Binary Call Option & Binary Put Option



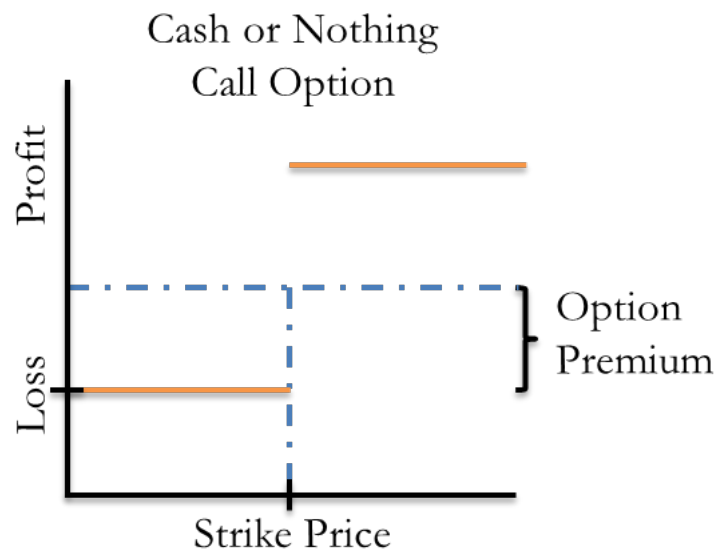
Source: Code & Finance, 2013

Figure 2. Binary Call Option vs. Vanilla Call Option



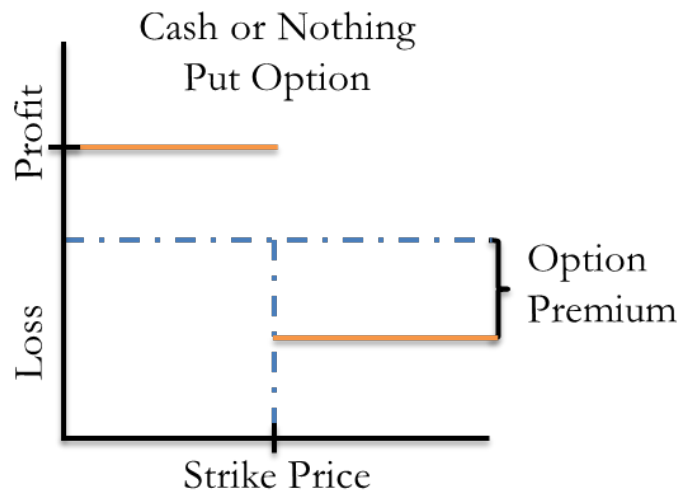
Source: EarnForex, 2021

Figure 3. Cash-or-Nothing Call Option



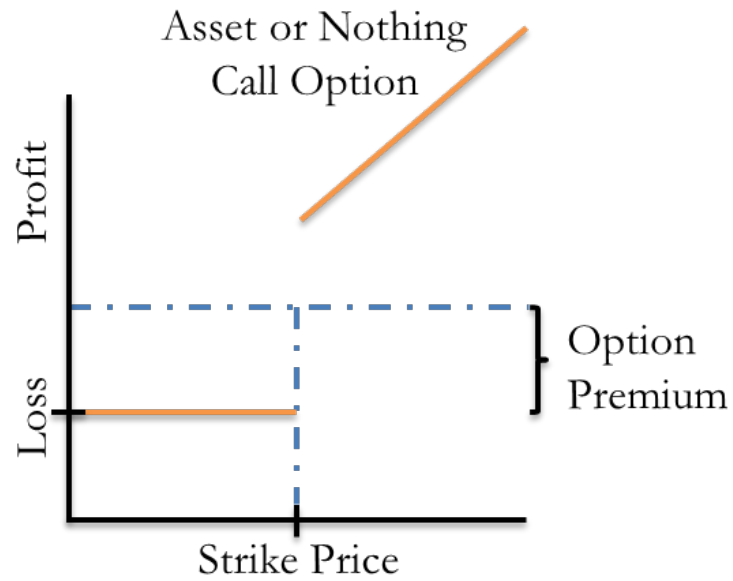
Source: SLCG, 2013

Figure 4. Cash-or-Nothing Put Option



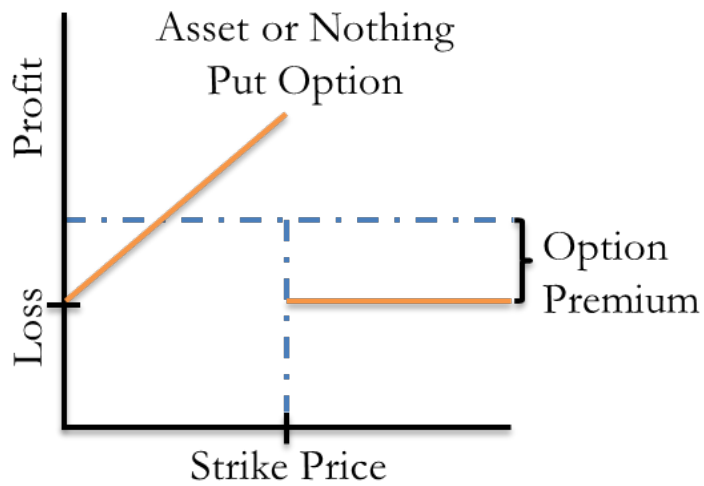
Source: SLCG, 2013

Figure 5. Asset-or-Nothing Call Option



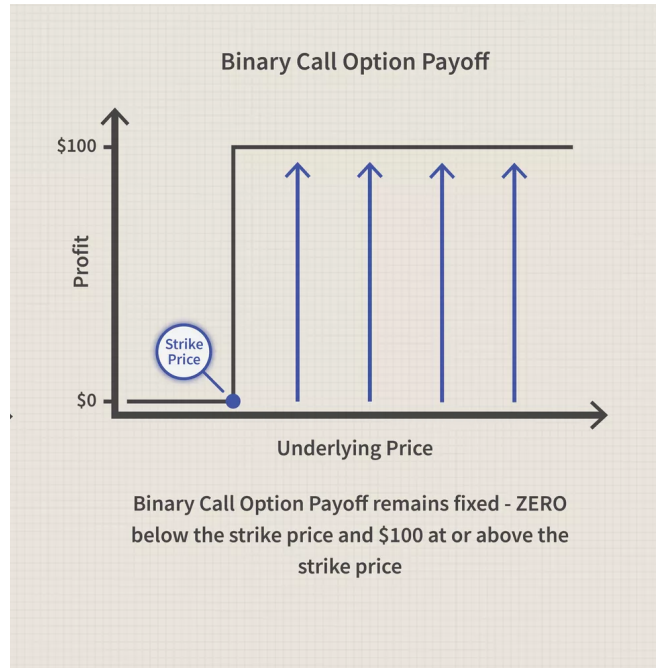
Source: SLCG, 2013

Figure 6. Asset-or-Nothing Put Option



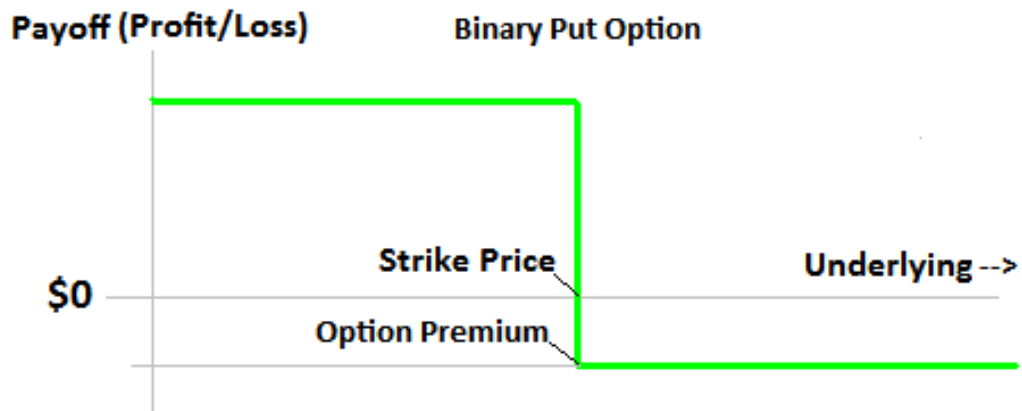
Source: SLCG, 2013

Figure 7. Profit/Payoff of Binary Call Option



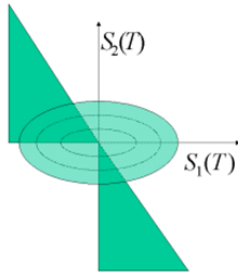
Source: Investopedia, 2021

Figure 8. Profit/Payoff of Binary Put Option



Source: Binary Options Blogger, 2021

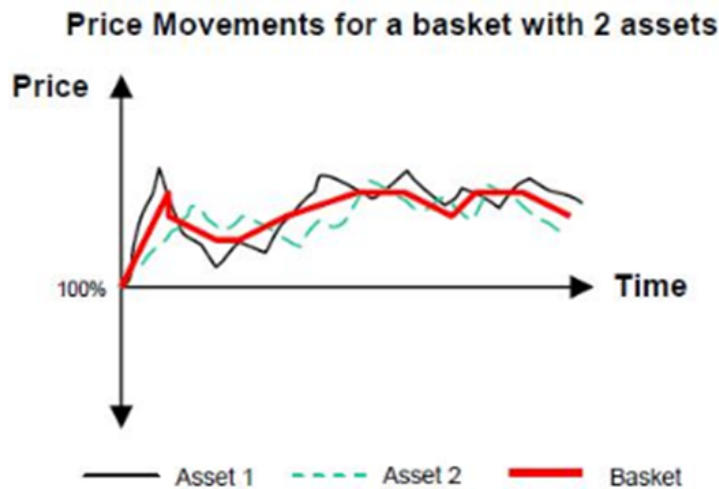
Figure 9. Example of a Currency Basket Option



Source: FX Basket Options - Jürgen Hakala, Uwe Wystup, 2008

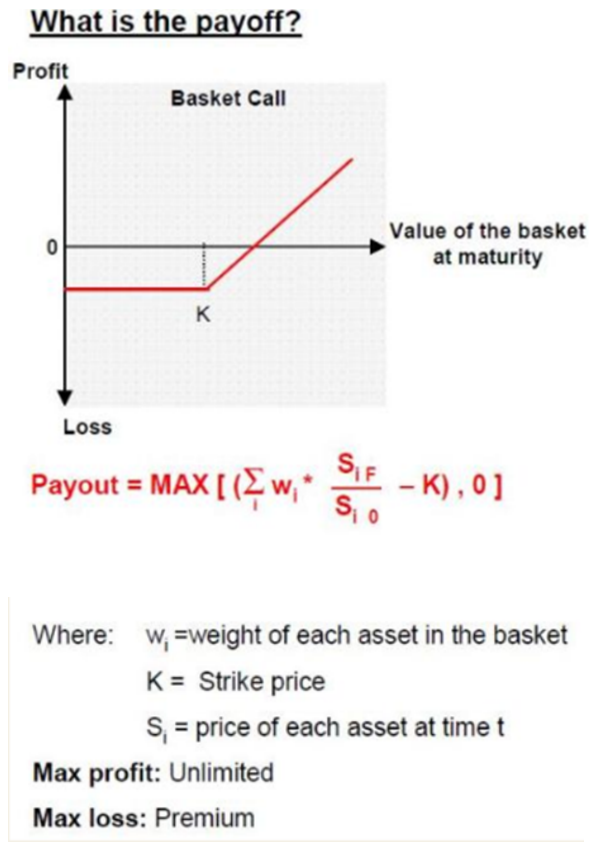
A basket option protects against a drop in both currencies at the same time. Individual options on each currency cover some cases, which are not protected by a basket option (shaded triangular areas) and that is why they cost more than a basket. The ellipsoids connect the points that are reached with the same probability assuming that the forward prices are at the center.

Figure 10. Pricing Basket Options



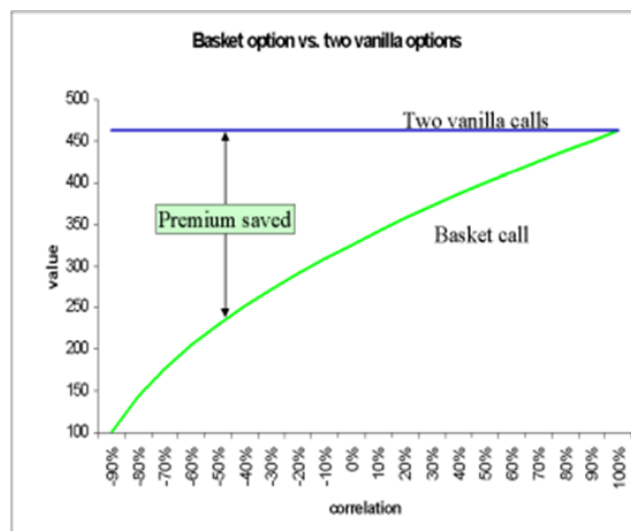
Source: Basket option | The Financial Engineer

Figure 11. Payoff of Basket Options



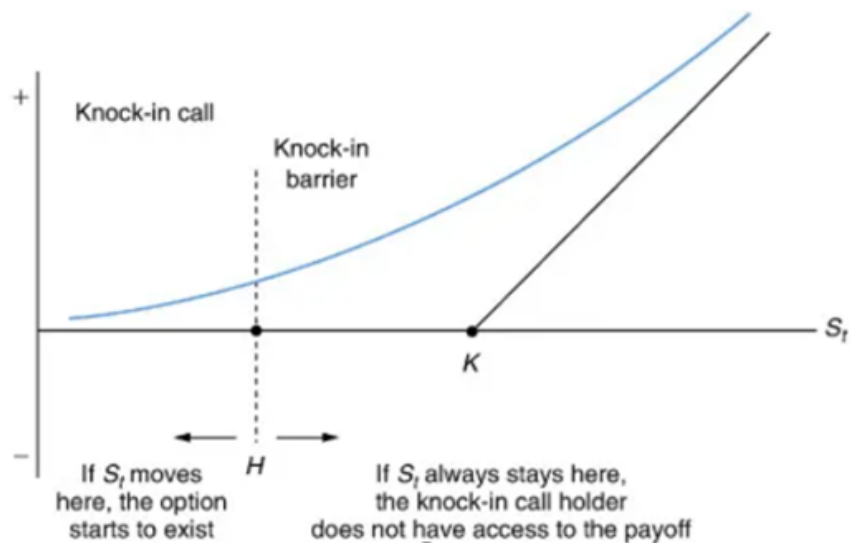
Source: Basket option | The Financial Engineer

Figure 12. Basket Option vs Two Vanilla Options



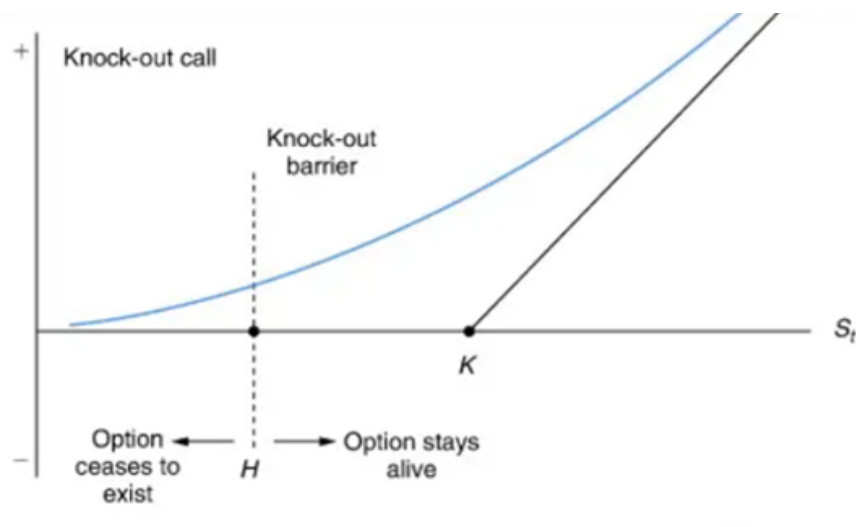
Source: <https://papers.ssrn.com>

Figure 13. Barrier Options: Knock-in call



Source: <https://corporatefinanceinstitute.com/resources/knowledge/trading-investing/barrier-option/>

Figure 14. Barrier Options: Knock-out call



Source: <https://corporatefinanceinstitute.com/resources/knowledge/trading-investing/barrier-option/>